Homework #2

1. 通过时谐场量的时域形式和频域形式的转换关系式推导出平均玻印亭矢量，并定义复数玻印亭矢量。

解. $S(r,t)=E(r,t)\times H(r,t)$

$$E(r,t)=Re\{E(r)e^{j\omega t}\}=\frac{1}{2}\{E(r)e^{j\omega t}+E^*(r)e^{-j\omega t}\}$$

$$H(r,t)=Re\{H(r)e^{j\omega t}\}=\frac{1}{2}\{H(r)e^{j\omega t}+H^*(r)e^{-j\omega t}\}$$

$$E(r,t)\times H(r,t)=\frac{1}{4}[E(r)\times H(r)e^{j2\omega t}+E(r)\times H^*(r)+E^*(r)\times H(r)+E^*(r)\times H^*(r)e^{-j2\omega t}]$$

$$S_{avg}=\frac{1}{T}\int_0^T S(r,t)dt=\frac{1}{T}\int_0^T[E(r,t)\times H(r,t)]dt$$

$$=\frac{1}{4T}\int_0^T[E(r)\times H^*(r)+E^*(r)\times H(r)]dt$$

$$\Rightarrow S_{avg}=\frac{1}{4T}[E(r)\times H^*(r)+E^*(r)\times H(r)]=\frac{1}{2}Re\{E(r)\times H^*(r)\}$$

故定义 $\overline{S_c}=\frac{1}{2}(\overline{E}\times\overline{H^*})$，则 $\overline{S_{avg}}=Re[\overline{S_c}]$

2. 推导出使用远场区的电场表示的辐射强度

解. 远场区，$S_{avg}=\frac{1}{2}Re[\overline{E}\times\overline{H^*}]$

$$=\frac{1}{2}Re[\overline{E}\times\frac{\hat{r}\times\overline{E}^*}{\eta}]=\hat{r}\frac{|\overline{E}|^2}{\eta}$$

故 $U(\theta,\phi)=\frac{S_{avg}\cdot dA}{d\Omega}=r^2S_{avg}$

故 $U(\theta,\phi)=r^2\frac{|\overline{E}|^2}{2\eta}$

T2.36, 2.53, 2.54, 2.58, 2.69, 2.71, 2.94

2.36. A linearly polarized wave traveling in the positive z-direction is incident upon a circularly polarized antenna. Find the polarization loss factor PLF (dimensionless and in dB) when the antenna is (based upon its transmission mode operation)
(a) right-handed (CW) (b) left-handed (CCW)

解: $PLF = |\hat{\rho}_w \cdot \hat{\rho}_a|^2$, 不妨令 $\hat{\rho}_w = \hat{a}_x$

(a) for right-handed, $\hat{\rho}_a = \frac{1}{\sqrt{2}}(\hat{a}_\theta + j\hat{a}_\phi) \Rightarrow \hat{\rho}_a^* = \frac{1}{\sqrt{2}}(\hat{a}_\theta - j\hat{a}_\phi)$

则 $PLF = |\hat{a}_x \cdot \frac{1}{\sqrt{2}}(\hat{a}_\theta - j\hat{a}_\phi)|^2 = \frac{1}{2}|\hat{a}_x \cdot \hat{a}_\theta - j\hat{a}_x \cdot \hat{a}_\phi|^2$

$\hat{a}_\theta = \hat{a}_x \cos\theta\cos\phi + \hat{a}_y \cos\theta\sin\phi - \hat{a}_z \sin\theta$

$\hat{a}_\phi = -\hat{a}_x \sin\phi + \hat{a}_y \cos\phi$

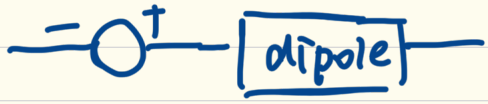
$\Rightarrow PLF = \frac{1}{2}(\cos^2\theta\cos^2\phi + \sin^2\phi)$

(b) for left-handed, $\hat{\rho}_a = \frac{1}{\sqrt{2}}(\hat{a}_\theta - j\hat{a}_\phi) \Rightarrow \hat{\rho}_a^* = \frac{1}{\sqrt{2}}(\hat{a}_\theta + j\hat{a}_\phi)$

则 $PLF = |\hat{a}_x \cdot \frac{1}{\sqrt{2}}(\hat{a}_\theta + j\hat{a}_\phi)|^2 = \frac{1}{2}|\hat{a}_x \cdot \hat{a}_\theta + j\hat{a}_x \cdot \hat{a}_\phi|^2 = \frac{1}{2}(\cos^2\theta\cos^2\phi + \sin^2\phi)$

2.53. A $\lambda/2$ dipole, with a total loss resistance of 1 ohm, is connected to a generator whose internal impedance is $50 + j25$ ohms. Assuming that the peak voltage of the generator is 2 V and the impedance of the dipole, excluding the loss resistance, is $73 + j42.5$ ohms, find the power
(a) supplied by the source (real) (b) radiated by the antenna
(c) dissipated by the antenna

解:



$Z_g = (50 + j25)\Omega$ $R_r = 73\Omega$ $X_A = 42.5\Omega$ $R_L = 1\Omega$ $Z_{in} = (74 + j42.5)\Omega$

$$I_g = \frac{V_g}{Z_g + Z_{in}} = \frac{2}{124 + j67.5} = (12.44 - 6.77j) \times 10^{-3} A = 14.17 \times 10^{-3} \angle -28.56° A$$

(a) $P_s = \frac{1}{2}\mathrm{Re}(V_g \cdot I_g^*) = 12.44 \times 10^{-3} W = 12.44 mW$

(b) $P_r = \frac{1}{2}|I_g|^2 R_r = 7.33 \times 10^{-3} W = 7.33 mW$

(c) $P_L = \frac{1}{2}|I_g|^2 R_L = 0.10 \times 10^{-3} W = 0.10 mW$

2.54. The antenna and generator of Problem 2.53 are connected via a 50-ohm $\lambda/2$-long lossless transmission line. Find the power

(a) supplied by the source (real) (b) radiated by the antenna

(c) dissipated by the antenna

解



$V_{peak}=2V$ $Z_0=50\Omega$ $R_r=73\Omega$ $X_A=42.5\Omega$

$Z_g=(50+j25)\Omega$ $l=\frac{\lambda}{2}$ $R_L=1\Omega$

$Z_{load}=(74+j42.5)\Omega$

$$\beta l=\frac{2\pi}{\lambda}\frac{\lambda}{2}=\pi$$

$$Z_{in}'=Z_0\frac{Z_{load}+jZ_0\tan(\beta l)}{Z_0+jZ_{load}\tan(\beta l)}=Z_{load}$$

故同2.53

(a) $P_{supplied}=12.44mW$ (b) $P_{rad}=7.33mW$ (c) $P_{diss}=0.10mW$

2.58. A dipole antenna consists of a circular wire of length l. Assuming the current distribution on the wire is cosinusoidal, i.e.,

$$I_z(z)=I_0\cos\left(\frac{\pi}{l}z'\right) \quad -l/2\le z'\le l/2$$

where I_0 is a constant, derive an expression for the loss resistance R_L which is one-half of (2-90b).

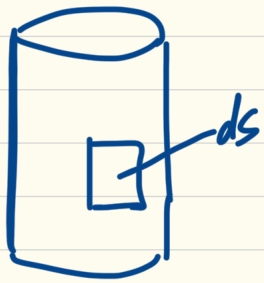
解 $H_\phi(r=a)=\frac{I_0}{2\pi a}\cos(\frac{\pi}{l}z')$

$$dw=\frac{1}{2}|H_\phi|^2R_s\,ds=\frac{1}{2}|H_\phi|^2R_s\,a\,d\phi\,dz$$

$$dW=\left(\frac{I_0}{2\pi a}\right)^2\frac{R_s}{2}\cos^2(\tfrac{\pi}{l}z)\,a\,d\phi\,dz$$

$$W=\int_{-\frac{l}{2}}^{\frac{l}{2}}\int_0^{2\pi}\frac{I_0^2R_s}{8\pi^2a^2}\cos^2(\tfrac{\pi}{l}z)\,a\,d\phi\,dz$$

$$=\frac{I_0^2}{4\pi}\frac{l\cdot R_s}{a}\frac{1}{2}=\frac{1}{2}I_0^2R_L \Rightarrow R_L=\frac{lR_s}{4\pi a}$$



2.69. The field radiated by an infinitesimal dipole of very small length ($l \leq \lambda/50$), and of uniform current distribution I_o, is given by (4-26a) or

$$\mathbf{E} = \hat{\mathbf{a}}_\theta E_\theta \approx \hat{\mathbf{a}}_\theta j\eta \frac{kI_o l}{4\pi r} e^{-jkr} \sin\theta$$

Determine the

(a) vector effective length

(b) maximum value of the vector effective length. Specify the angle.

(c) ratio of the maximum effective length to the physical length l.

解. (a) $E_{rad} = -j\frac{\eta k I_{in}}{4\pi r} e^{-jkr} l_e(\theta,\phi)$

对比可得, $l_e = -\hat{a}_\theta l \sin\theta$

(b) $|l_e| = l\sin\theta$ $\quad |l_e|_{max} = l$, 此时 $\theta = 90^\circ$

(c) $\frac{|l_e|_{max}}{l} = 1$

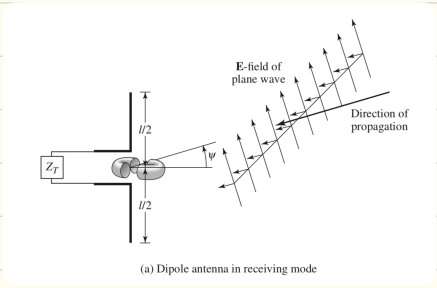
2.71. A uniform plane wave, of 10^{-3} watts/cm^2 power density, is incident upon an infinitesimal dipole of length $l = \lambda/50$ and uniform current distribution, as shown in Figure 2.29(a). For a frequency of 10 GHz, determine the maximum open-circuited voltage at the terminals of the antenna. See Problem 2.69.

解. $W = \frac{1}{2\eta_0}|E|^2$ $\quad \lambda = \frac{c}{f}$

$\Rightarrow |E| = \sqrt{2\eta_0 W}$ $\quad \Rightarrow l = \frac{\lambda}{50} = \frac{c}{50f}$

由 Prob 2.69, $|l_e|_{max} = l$

$V_{oc}|_{max} = |E| \cdot |l_e|_{max} = \sqrt{2\eta_0 W}\,\frac{c}{50f} = 52.1\,mV$



(a) Dipole antenna in receiving mode

2.94. A communication satellite is in stationary (synchronous) orbit about the earth (assume altitude of 22,300 statute miles). Its transmitter generates 8.0 W. Assume the transmitting antenna is isotropic. Its signal is received by the 210-ft diameter tracking paraboloidal antenna on the earth at the NASA tracking station at Goldstone, California. Also assume no resistive losses in either antenna, perfect polarization match, and perfect impedance match at both antennas. At a frequency of 2 GHz, determine the:

(a) power density (*in watts/m*^{2}) incident on the receiving antenna.

(b) power received by the ground-based antenna whose gain is 60 dB.

解. 22300 statute miles = 3.589×10^7 m

(a) $P_i = \frac{P_{rad}}{4\pi R^2} = 4.94 \times 10^{-16}\ W$

(b) $D_0 = 60\ dB = 10^6$ $\quad \lambda = \frac{c}{f} = 0.15m$

$A_{em} = \frac{\lambda^2}{4\pi} D_0$

$P_r = A_{em} P_i = \frac{\lambda^2}{4\pi} D_0 P_i = 8.85 \times 10^{-13}\ W$

4. Given

$$\underline{E}^a = \hat{a}_\theta E_\theta^a + \hat{a}_\phi E_\phi^a = \left(-j\eta \frac{kI_{in}}{4\pi r} e^{-jkr}\right) \underline{\ell}_e(\theta,\phi) \quad (1)$$

and

$$U(\theta,\phi) = r^2 W_{rad}(\theta,\phi) = \frac{r^2 \left|\underline{\boldsymbol{E}}^a(\theta,\phi)\right|^2}{2\eta} \quad (2)$$

derive

$$l_e^2(\theta,\phi) = \frac{4\pi R_r}{\eta k^2} D(\theta,\phi)$$

where Rr is the radiation resistance.

解. 将(1)式代入(2)式，得

$$U(\theta,\phi) = r^2 W_{rad}(\theta,\phi) = \frac{r^2}{2\eta}\left|-j\frac{\eta k I_{in}}{4\pi r} e^{-jkr} \underline{l}_e(\theta,\phi)\right|^2$$

$$= \frac{r^2}{2\eta} \frac{\eta^2 k^2 I_{in}^2}{16\pi^2 r^2} l_e^2(\theta,\phi)$$

$$= \frac{\eta k^2 |I_e|^2}{32\pi^2} l_e^2(\theta,\phi)$$

$$P_{rad} = \frac{1}{2}|I_e|^2 R_r$$

$$\Rightarrow D(\theta,\phi) = \frac{4\pi U(\theta,\phi)}{P_{rad}} = \frac{4\pi}{\frac{1}{2}|I_e|^2 R_r} \frac{\eta k^2 |I_e|^2}{32\pi^2} l_e^2(\theta,\phi)$$

$$= \frac{\eta k^2}{4\pi R_r} l_e^2(\theta,\phi)$$

$$\Rightarrow l_e^2(\theta,\phi) = \frac{4\pi R_r}{\eta k^2} D(\theta,\phi)$$